SQL Fundamentals.

01 -> SELECT *
FROM employees;

ID	First_Name	Last_Name	Department	Salary	Hire_Date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-03	Johannesburg

02 SELECT DISTINCT department
FROM employees;

Department
IT
HR
Finance
Marketing

03 -> SELECT first_name, last_name
FROM employees
ORDER BY salary DESC

First_name	Last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Tshegofatso Mogashi

SQL Exercise
CASE Statement

①

```
SELECT product_name, price;
        CASE
        WHEN price > 1000 THEN 'Expensive'
        WHEN price BETWEEN 100 AND 1000
        THEN 'Mid-range'
        END AS price_category
```

Product_name	price	price_category
laptop	1200	Expensive
Phone	800	Midrange
Keyboard	49	Affordable
Monitor	300	Mid-range
Mouse	25	Affordable

②

```
SELECT customer_name, amount
        CASE
            WHEN amount >=1000 THEN 'High Value'
            WHEN amount BETWEEN 500 AND
                 999.99 THEN 'Medium Value'
            WHEN amount < 500 THEN 'Low Value'
        END AS order_value_category
FROM orders.
```

Customer_name	amount	order_value_category
Alice	150.00	Low value
Bob	560.00	~~High~~ Medium value
Charlie	999.99	Medium value
Diana	45.50	Low Value
Ethan	1200.00	High Value

(3)

```
SELECT emp_name, department, salary,
    CASE
        WHEN department = 'IT' AND salary
        > 80 000 THEN 'Senior IT'
        WHEN department = 'HR' AND salary
        > 55 000 THEN 'Experience HR'
        ELSE 'staff'
    END AS position_level
FROM employees;
```

~~emp_name~~ ~~department~~ ~~sl~~

~~emp_name~~	department	salary	position_level
John	IT	85 000	Senior IT
Sara	HR	60 000	Experience HR
Mark	IT	75 000	staff
Lucy	Finance	95000	staff
Tom	HR	55000	staff

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④

```
SELECT student_name, score,
    CASE
      WHEN score >= 90 THEN 'A'
      WHEN score >= 80 THEN 'B'
      WHEN score >= 70 THEN 'C'
      WHEN score >= 60 THEN 'D'
    END AS '7'
FROM students;
```

student_name	score	grade
Anna	92	A
Ben	76	C
Cora	59	7
David	83	B
Ella	68	D

⑤

```
SELECT delivery_id, delivery_time_minutes,
    CASE
      WHEN delivery_time_minutes <= 30 THEN
      'fast'
      WHEN delivery_time_minutes <=60 THEN
      'on time'
      ELSE 'late'
    END AS performance
FROM delivery deliveries
```

delivery_id	delivery_time_minutes	performance
1	45	On time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

(06)

issue type ~~priority~~ ~~priority_label~~

login issue

~~Server down~~

(06)

```
SELECT issue_type, priority,
       CASE priority
         WHEN 3 THEN 'High'
         WHEN 2 THEN 'Medium'
         WHEN 1 THEN 'low'
       END AS priority_label
FROM tickets;
```

issue type	Priority	Priority_label
Login issue	1	low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

(07)

```
SELECT student_id,
       days_present * 100.0% / total days AS
       ~~perce~~ attendance_percentage
    CASE
      WHEN days_present * 100% / total_days >= 90
      THEN 'Excellent'
      WHEN days_present * 100% / total_days >= 75
      THEN 'Good'
    END AS attendance_status
FROM attendance;
```

Student_id	Attendance_percentage	Attendance status ~~Status~~
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

(08)

```
SELECT product_id, stock_qty,
    CASE
      WHEN stock_qty = 0 THEN 'Out of stock'
      WHEN stock_qty BETWEEN 1 AND 5
      THEN 'Low stock'
      WHEN stock_qty > 5 THEN 'In stock'
    END AS stock_status
FROM products_inventory;
```

product_id	stock_qty	stock_status
1	5	Low stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

(09)

```
SELECT subject, enrolled_students,
       CASE
           WHEN enrolled_students >= 25 THEN
           'large'
           WHEN enrolled_students >= 10 THEN
           'medium'
       ELSE 'small'
       END AS class_size_category
FROM classes;
```

subject	enrolled_student	class_size_
Math	30	large
English	25	large
Science	15	Medium
Art	5	Small
History	20	Medium.

(10)

```
SELECT payment_id, payment_method, amount,
    CASE
        WHEN payment_method = 'cash AND
        amount >= 200 THEN 'Eligible for discount'
        ELSE 'not Eligible'
    END AS discount_eligibility
FROM payments
```

payment_id	payment_method	amount	discount eligibility
1	Card	50	Not eligible
2	Cash	200	Eligible for discount
3	Card	180	Not Eligible
4	Paypal	75	Not Eligible
5	Cash	300	Eligible for discount